GAEB Modern Platform

From AS/400 to a governed, Arabic-first platform

Executive presentation — the customer pilot proposal · Veebase · All results shown are verified engineering results on synthetic data

Why we are here

- GAEB's six business systems still run on IBM i (AS/400) V6R1 — reliable, but out of support and hard to staff
- The risk is not the hardware: it is meaning — Arabic records in legacy codepages, business rules that exist only inside programs
- Veebase has built the replacement platform, tested it at scale, and rehearsed migration end to end on synthetic data
- Today we propose the next step only: a governed pilot on GAEB's own data, under GAEB's written authorization

The legacy estate — what the evidence shows

- 502 legacy program and file objects across six subsystems
- 1,068 fields mapped in the reconciliation register
- One money path links the systems: estimate → certificate → budget payment
- Two file definitions conflict between copies; some masters were never delivered — stated openly, gated honestly
- Reference exports hit legacy row caps — true volumes to be confirmed by GAEB's DSPFD counts

What we built

- A standalone bilingual platform — no dependency on outside infrastructure
- Six complete business workflows, not data-entry screens
- An offline migration engine that never connects to, and never writes to, the AS/400
- An authorization system that refuses customer data until GAEB's approvals exist as verified records

Platform architecture

Presentation

Arabic / English web interface — full RTL, tested dictionary parity (350+ keys per language)

API & governance

Role-based permissions, revocable sessions, login throttling, strict security headers, append-only audit

Data

PostgreSQL 16 — exact decimal money end to end; no floating point, no silent rounding

Migration

Offline staging with per-record lineage and original bytes; field-level Arabic codepage decoding

Six modules, complete workflows

School Map

Building registry, server-side search at million-record scale

Budget

Allocations and execution — money visible only to finance roles

Tenders & Contracts

A 12-stage lifecycle machine — stages cannot be skipped

Payment Certificates

Exact valuation; the certifier cannot authorize payment

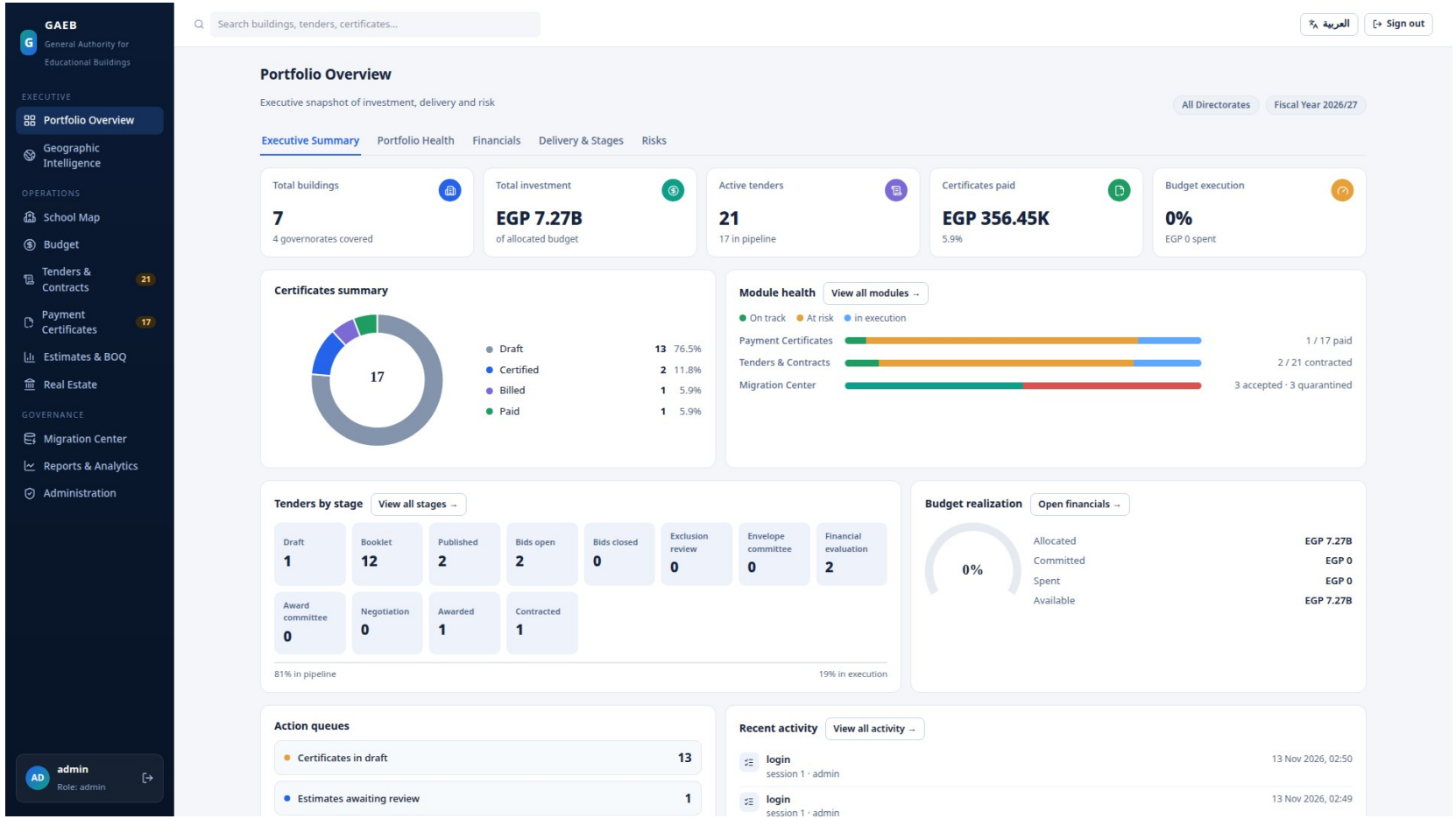
Estimates & BOQ

Item catalogue, revisions with full history

Real Estate

Expropriation cases bound to the ministerial-decree gate

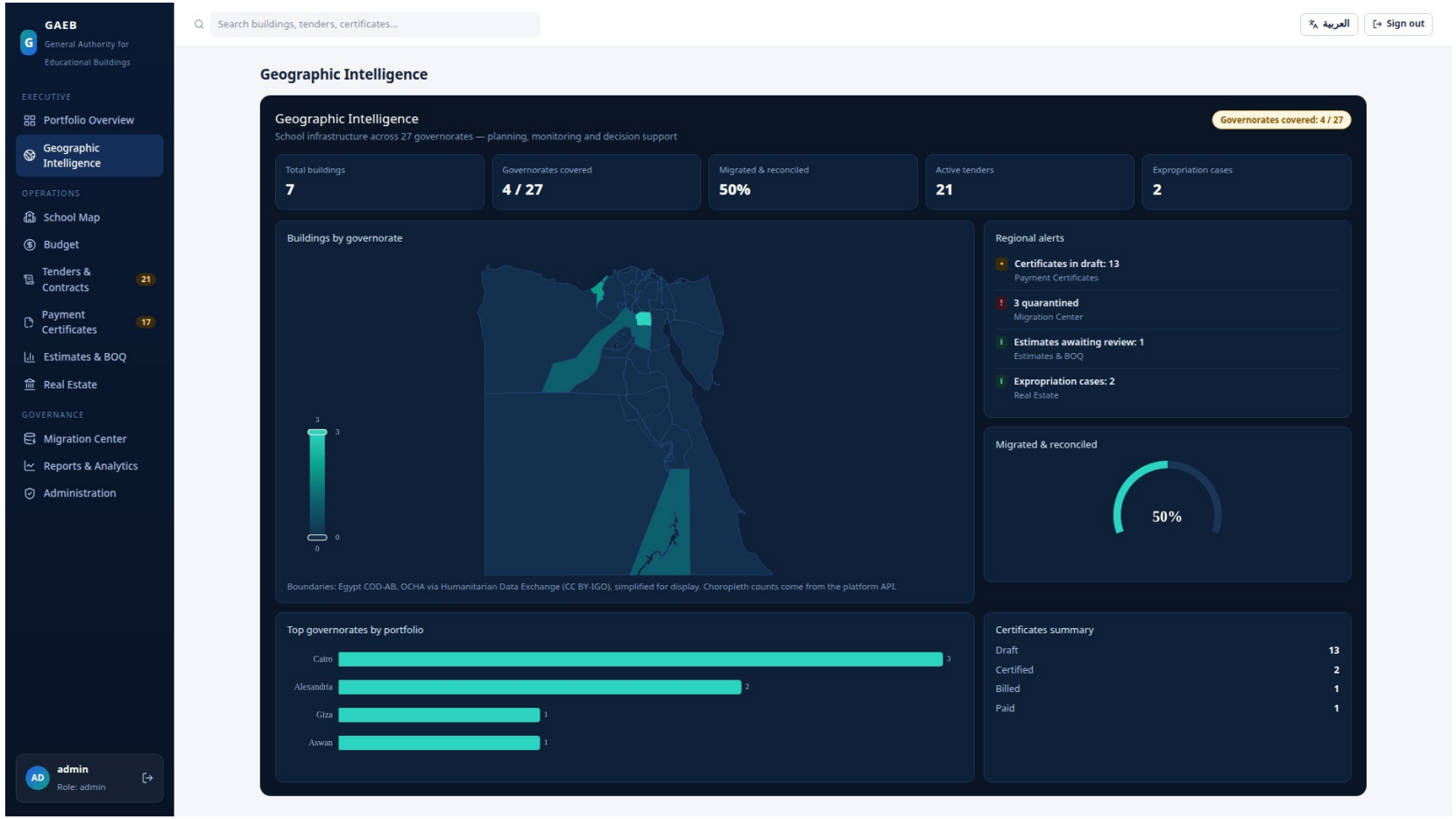
Executive Command Center



Live build — synthetic demonstration data

- Every figure is computed live from platform data
- Action queues show work waiting for a human decision
- Finance figures appear only for finance-permission holders

Geographic Intelligence



- GAEB legacy codes joined to international P-codes
- Coverage reported honestly — migrated vs not yet
- One click from a governorate to its buildings

Official 27-governorate boundaries (OCHA COD-AB, CC BY-IGO) — synthetic data

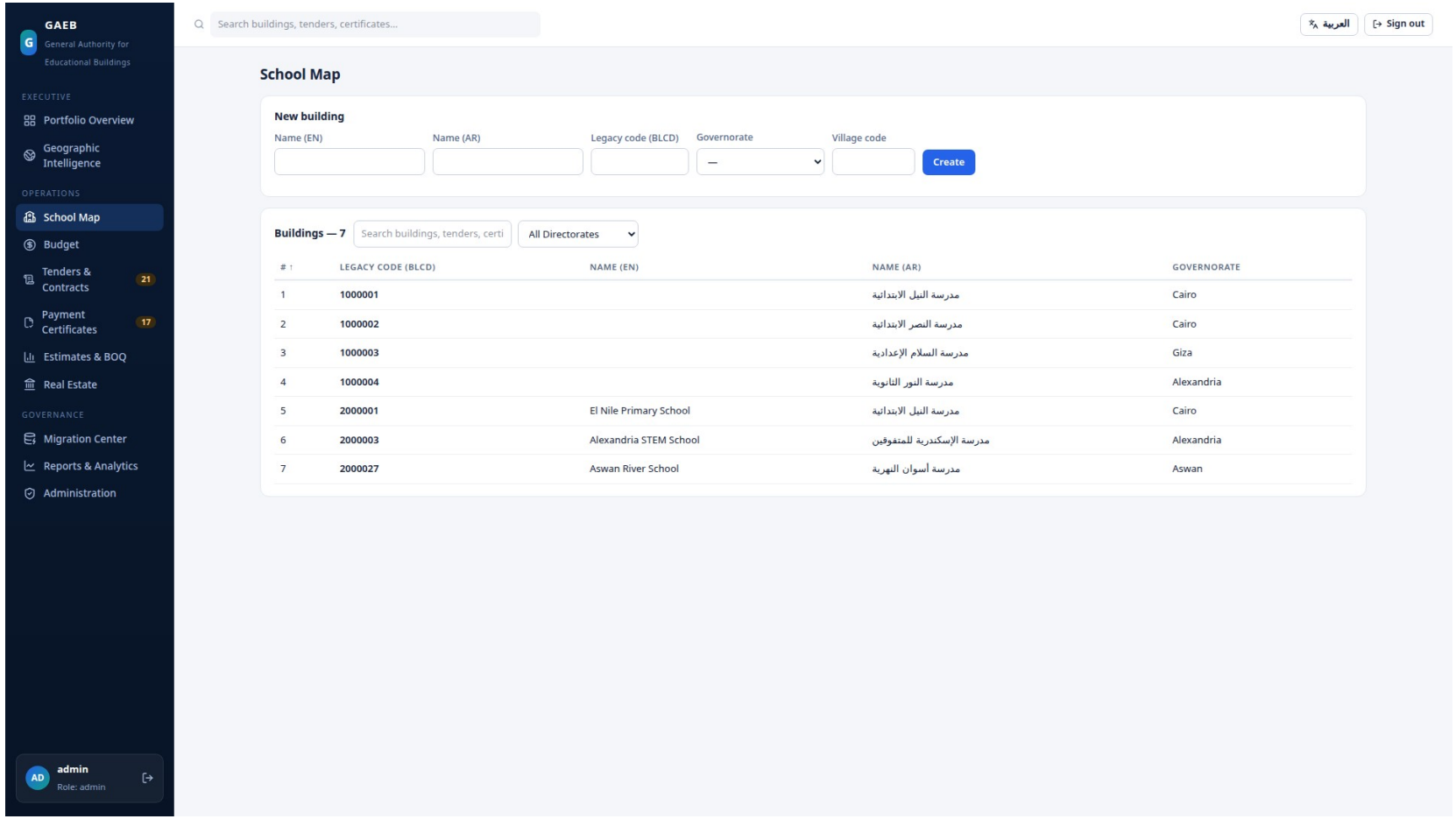
Arabic-first, not Arabic-translated



The same live data in Arabic — full RTL

- Field-level decoding: IBM 420/864, Windows-1256
- Originals preserved — search normalization never overwrites
- Unsupported codepages stop for a decision, never a guess

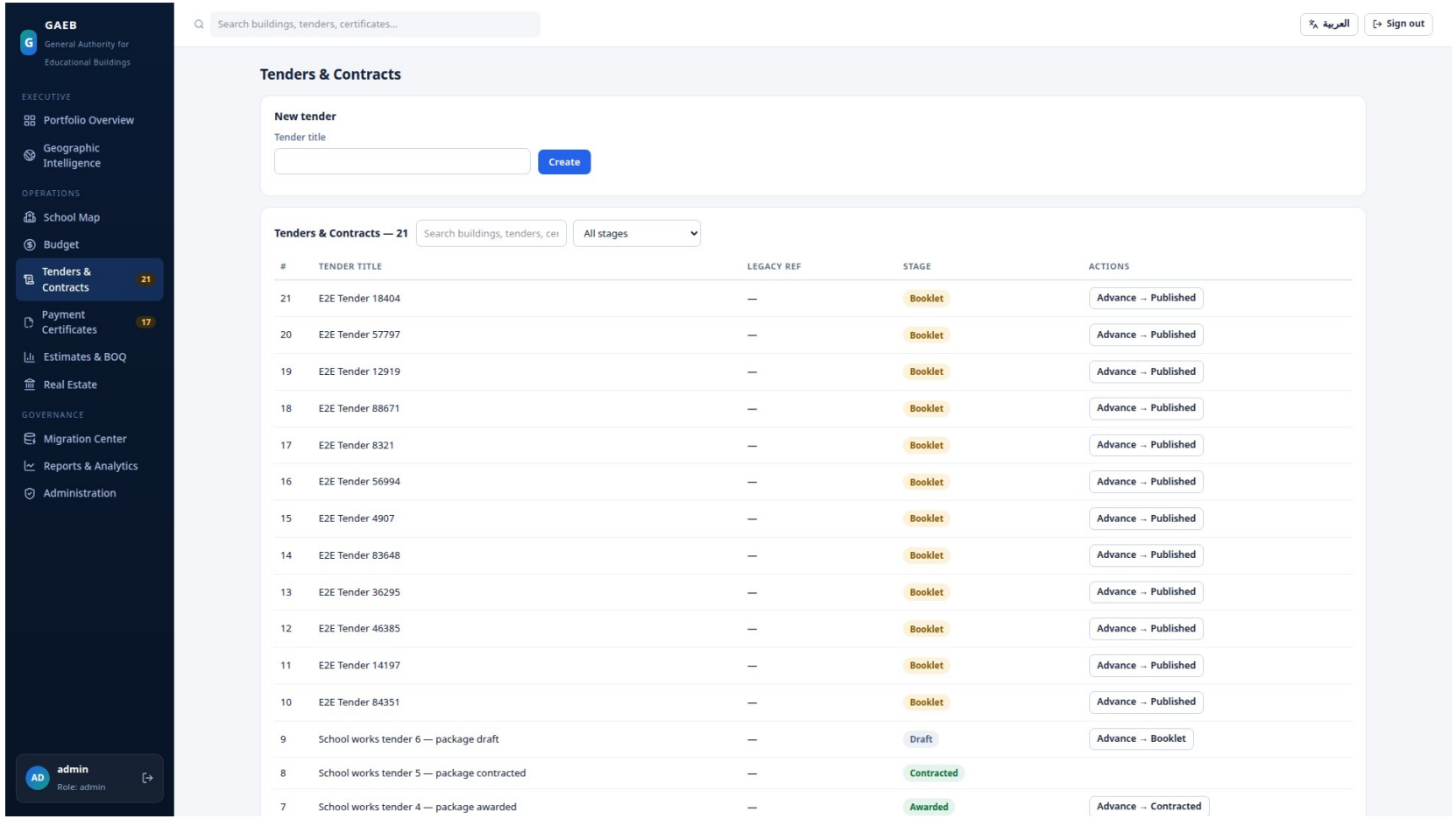
School Map — the building registry



- Server-side search and pagination — validated at 1,000,000 synthetic records
- The legacy BLCD code is preserved as the lineage key on every building

Synthetic demonstration data

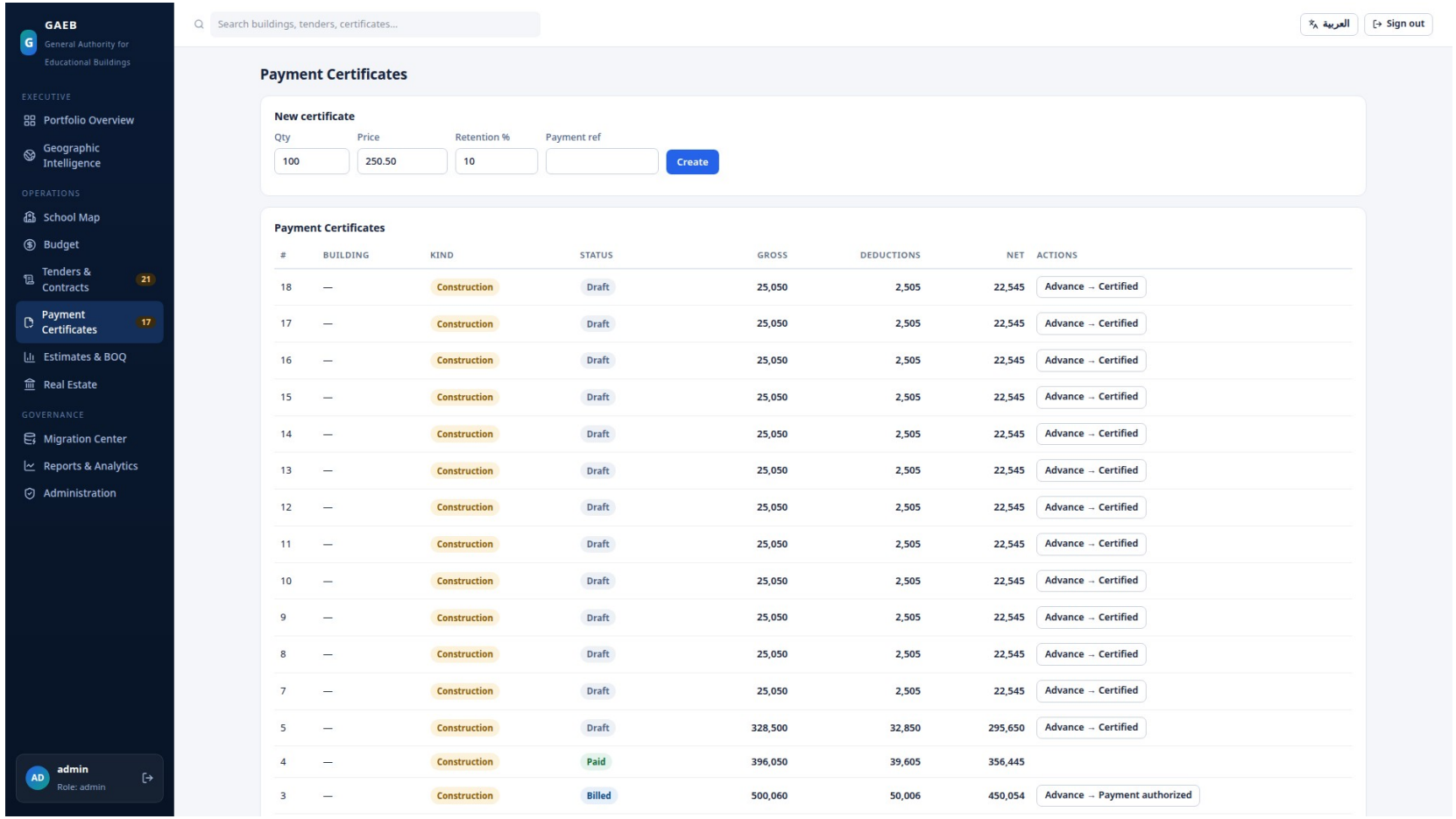
Tenders — a lifecycle, not a form



- 12 stages from draft to contracted — skipping is refused by the system
- Every stage transition is written to history

Synthetic demonstration data

Certificates — money with separation of duties



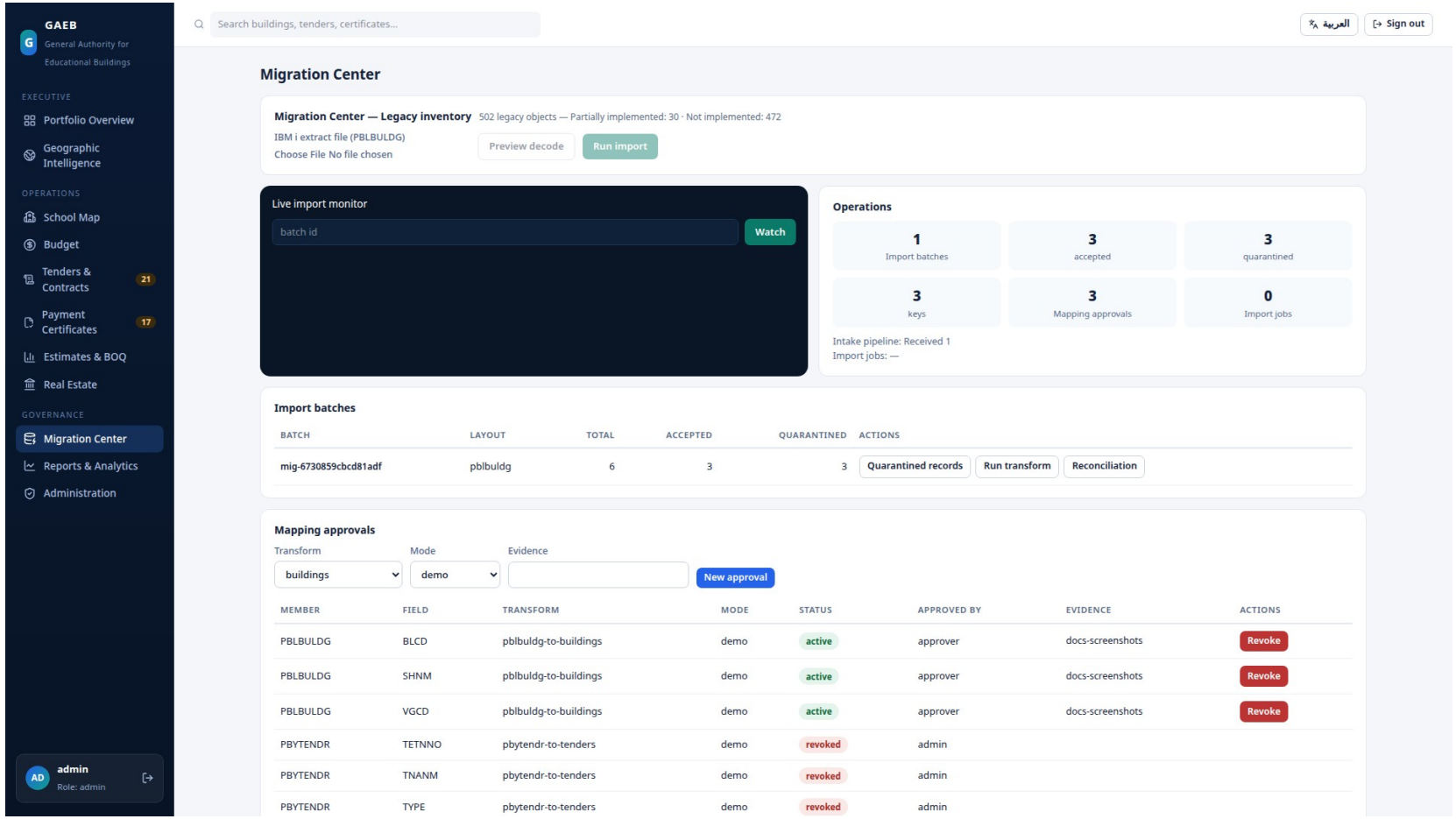
- Valuation is exact: quantity × price less typed deductions
- The user who certifies a certificate cannot authorize its payment — the server refuses

Synthetic demonstration data

Migration — controlled in 15 steps

- Written authorization → classification → approved extraction at GAEB
- SHA-256 at source and at receipt; a mismatch halts intake
- Encoding preflight → tracked import → reconciliation → customer review
- Retention or deletion only per GAEB's approved policy
- Customer data never enters the demonstration environment — and is never sent to any external or AI service

Reconciliation — every record accounted for



A real (synthetic) batch in the Migration Center

- The invariant: source = loaded + quarantined
- Quarantined records carry their reason and their raw bytes — nothing is silently dropped
- Money control totals are exact decimals, end to end

Authorization governance — evidence, not assertions

- Every required approval is an evidence record with a lifecycle: pending → submitted → verified — then revocable, expirable, supersedable
- Separation of duties: whoever submits evidence can never verify it — at any role, administrators included
- A typed-in document hash is labeled unverified; only a server-computed hash from the actual bytes is labeled verified
- We do not claim to authenticate GAEB signatures cryptographically — verification is an independent human decision, recorded under their identity

Continuous authorization — valid now, not once

- Eligibility is recomputed from current state at every customer operation — import and transform alike
- Revoke an approval or an evidence record, and the next operation refuses — not the next activation
- Every import is bound to an authorized dataset and to the authorized file's hash — a substituted file is refused
- Streaming imports re-verify after every committed chunk; staged records can never become operational data unauthorized

The pilot — three weeks after entry conditions

- Week 0: host qualification by automated preflight, restore drill, alert test, accounts, evidence verification, activation
- Week 1: controlled intake — first extract, hash verification, first migration, reconciliation package #1
- Week 2: business validation — Arabic fidelity review, module UAT, defect triage and controlled re-runs
- Week 3: acceptance — final reconciliation, UAT retests, handover, pilot closure, production-gap assessment

Decisions we ask GAEB to make

- BD-07 — duplicate record policy (blocks the first import)
- BD-08 — historical migration scope (blocks pilot sizing)
- BD-01…BD-06 — module semantics, closed in the same one-day workshop
- Five approvals: charter, scope, extract authorization, environment, named participants
- Every decision is recorded in a register GAEB signs in the room — and enters the platform as independently verified evidence

Success — measured, not asserted

- 100% record accounting on every batch; the reconciliation equation exact
- Money control totals exact; Arabic fidelity 100% on the agreed sample
- Zero imports without a current authorization
- UAT critical path passed; every exception closed by a recorded decision
- These criteria are our proposal — they bind only when GAEB adopts them

Verification status — stated plainly

- Verified: 65 backend, 32 frontend, 40 browser, 31 visual tests — and a 34-check end-to-end rehearsal
- Verified: backup restore, physical restore, and point-in-time recovery
- Not yet: CI execution (one owner authorization), Docker certification, independent security review, screen-reader validation
- Not claimed anywhere: customer acceptance, production readiness

The ask

- A 30-minute executive meeting — this week or next — to fix the workshop date
- One day of your decision owners' time
- Signatures: charter, then the first extract authorization
- Then Week 0 begins — and the platform itself will hold the sequence, refusing every shortcut
- The first real batch reconciles under GAEB's own signature — or it does not ship